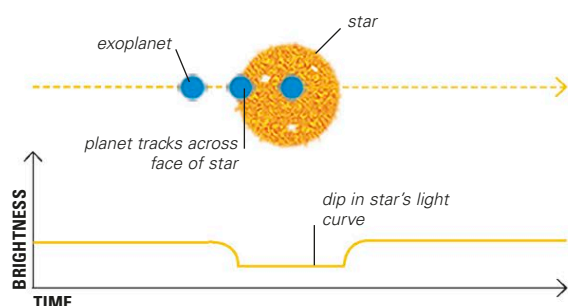


DETECTING EXTRASOLAR PLANETS

As extrasolar planets are invariably much smaller and dimmer than their parent stars, detecting them presents many challenges. Only about 30 or so have been found by direct imaging, which involves first blocking out the light from the parent star. All other exoplanet discoveries have been made by indirect methods. The most productive so far has been the Doppler spectroscopy or radial velocity method, which is based on the use of a sensitive instrument called a spectrograph. It relies on the fact that as an exoplanet orbits its parent star, its gravitational pull produces a tiny “wobble” in the star’s movement relative to Earth. A second indirect approach proving increasingly productive is the transit method, which involves looking for repeated slight dips in the brightness of a star as a planet passes in front of it. One advantage of this method is that it reveals the planet’s diameter. Several other indirect detection methods have also been employed, with varying



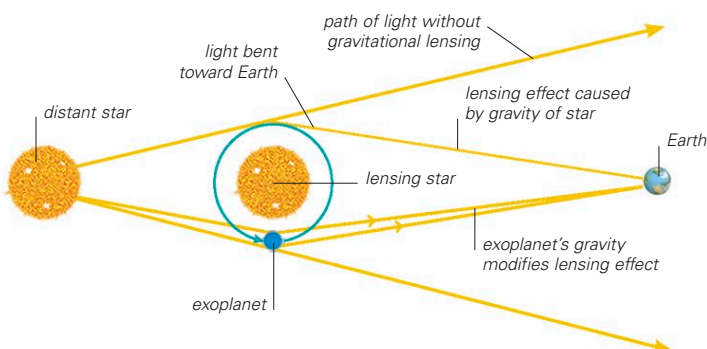
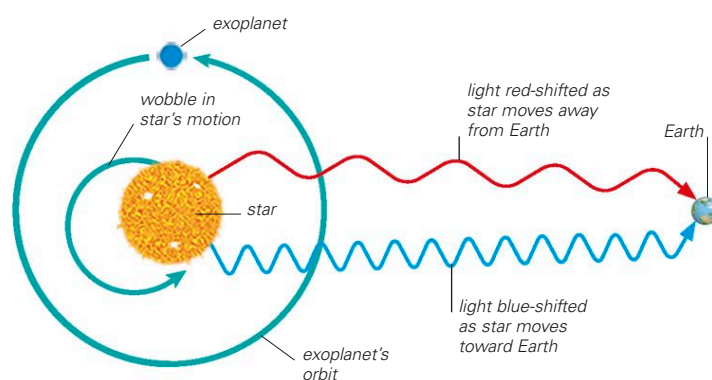
success. These include gravitational microlensing (detecting variations in the lensing effect of a star’s gravitational field, caused by a planet orbiting the star) and the pulsar timing method, which detects exoplanets orbiting pulsars from slight anomalies in the timing of the pulsars’ radio pulses.

THE TRANSIT METHOD

This approach involves observing repeated transits of a planet in front of its parent star. Each transit causes a slight dip in the star’s brightness—of the order of 0.01 percent for an Earth-size planet.

DOPPLER SPECTROSCOPY

An exoplanet’s orbit causes a “wobble” in the motion of its parent star. As a result, light waves coming from the star appear to be alternately slightly lengthened (red-shifted) and shortened (blue-shifted)—a phenomenon that can be detected by sensitive spectrographs (see p.33).

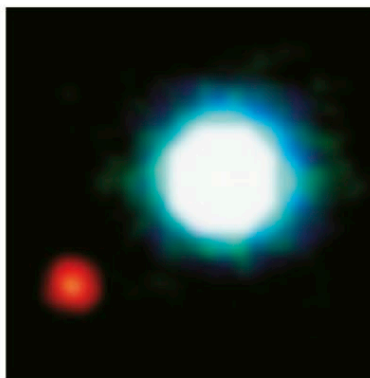


GRAVITATIONAL MICROLENSING

The gravitational field of a star acts like a lens that can bend light rays coming from a distant background star, thus magnifying that star as seen from Earth. The presence of an exoplanet orbiting the lensing star produces detectable variations in the degree of magnification, or lensing effect, over time.

DIRECT IMAGING

This composite image was made in 2004 with a telescope located at the Paranal Observatory in Chile. It shows a brown dwarf star (here appearing bright white) known as 2M1207 and a smaller red companion, thought to be a hot gas-giant planet. This red object is the first extrasolar planet ever to have been directly imaged.



MYSTERIOUS PULSAR PLANETS

The very first exoplanets to be discovered, in 1992, were not in solar systems around Sun-like stars, but instead orbited pulsar PSR B1257+12—the rapidly spinning burned-out remnant of a once-massive star (see p.267). So-called “pulsar planets” can be identified because a pulsar’s rate of rotation is constant over very long periods and can be measured with great precision. When the gravity of one or more planets pulls on the pulsar, it can create slow, regular changes in the pulsation rate. Three planets were discovered orbiting PSR B1257+12 (claims of a fourth are now disproved), the innermost of which—just twice the mass of the Moon—is the least massive planet known. The discovery was unexpected, since pulsars are formed during supernova explosions that would destroy any systems in orbit around them. Astronomers therefore think that pulsar planets are a second generation formed from debris left in orbit once the explosion subsides. Planets orbiting three more pulsars have since been confirmed, with several others proposed. In 2006, the Spitzer Space Telescope even discovered an apparently recent protoplanetary disk orbiting around a 100,000-year-old supernova.

SEARCHING FOR EXTRASOLAR PLANETS

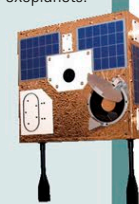
The organized search for exoplanets has a history going back to 1987. For each of the ongoing search programs, missions, or instruments listed below, the year in which it began operation is given.

1992 FIRST PULSAR PLANET Aleksander Wolczan and Dale Frail use the Arecibo radio telescope to discover two planets in orbit around the millisecond pulsar PSR B1257+12. A third smaller world is discovered in 1994.

1993 ELODIE/SOPHIE A spectrograph installed at an observatory in southeastern France, ELODIE discovered over 20 exoplanets—including (in 1995) the first to be found orbiting a Sun-like star. ELODIE was replaced by an improved instrument, SOPHIE, in 2006.

2002 MAGELLAN This program utilizes a spectrograph mounted on twin telescopes at the Las Campanas Observatory in Chile. By 2010, it had discovered nine exoplanets.

2003 MOST Canada’s first space telescope, MOST has been used to monitor giant exoplanets that transit their parent stars and to study atmospheric changes on the planets during the transits.

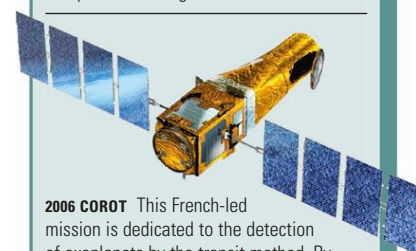


2003 SPITZER SPACE

TELESCOPE Spitzer has been used to study the light coming from exoplanets that transit their parent stars. In 2005, Spitzer made some of the first direct captures of infrared light from an exoplanet (although it did not resolve that light into actual images).



2003 HARPS A super-sensitive spectrograph at the European Southern Observatory in La Silla, Chile, HARPS has detected some 150 planets circling Sun-like stars.



2006 COROT This French-led mission is dedicated to the detection of exoplanets by the transit method. By the end of 2011, it had detected more than 20 new exoplanets.

2009 KEPLER This NASA mission used the transit method to study a single field of about 150,000 Milky Way stars, looking for Earth-sized exoplanets. From 2013 until its retirement in 2018, it carried out its “K2” follow-up mission with reduced sensitivity. In total, Kepler discovered 2,652 exoplanets.

2018 TESS SATELLITE This NASA satellite was designed to look for transits around bright nearby stars across almost the entire sky. In early 2020, TESS discovered TOI 700d, an Earth-sized world in the habitable zone around a red dwarf some 100 light-years away.